

Practical Lab 2: Data Preprocessing (Cleaning and Integration)

Objective: To perform practical data cleaning and integration tasks using the provided Student Data CSV files.

Required Files: student_details_batch1.csv, student_details_batch2.csv, and student_grades.csv.

Dataset Overview

Dataset Name	Description	Key Columns
student_details_batch1.csv & batch2.csv	Contains demographic and basic academic details of students. Contains missing values inserted specifically for imputation practice.	StudentID, Age, Monthly_Allowance, Department, Hostel_Status, Study_Hours
student_grades.csv	Contains academic performance metrics for a subset of the student population.	StudentID, CGPA, Credits_Completed

Task 1: Data Integration (Concatenation)

Before cleaning the data, we must combine our batches of student records into a single working dataset.

- **Step 1:** Load student_details_batch1.csv and student_details_batch2.csv using pandas.
- **Step 2:** Use pd.concat() to vertically stack both dataframes into a single dataframe called df_master_details.
- **Step 3:** Print the shape of the new dataframe using df_master_details.shape to verify all records are successfully combined.

```
import pandas as pd
# Your code here to load and concatenate the datasets
```

Task 2: Data Cleaning (Handling Missing Values)

Now, perform missing value imputation on the `df_master_details` dataframe. Use `df_master_details.isnull().sum()` to check missing value counts before and after performing each sub-task.

- **2.1 Mean Imputation (Age):** The Age column is generally normally distributed. Fill its missing values using the mean of the column.
- **2.2 Median Imputation (Monthly_Allowance):** The Monthly_Allowance column contains a few extreme outliers. Calculate the median and use it to fill the missing values to avoid skewed estimates.
- **2.3 Mode Imputation (Department):** Department is a categorical string column. Fill the missing values with the most frequently occurring department (the mode).
- **2.4 Constant Imputation (Hostel_Status):** Missing values in Hostel_Status indicate the student hasn't specified their living arrangement yet. Fill these missing values explicitly with the string "Not Specified".
- **2.5 KNN Imputation (Study_Hours):** Import `KNNImputer` from `sklearn.impute`. Create a subset dataframe with the numerical columns (Age, Monthly_Allowance, and Study_Hours) and use KNN Imputation (with `n_neighbors=5`) to estimate the missing Study_Hours based on a student's similarity to others. Replace the old Study_Hours column with the imputed values.

```
# Example for Median Imputation
```

```
df_master_details['Monthly_Allowance'].fillna(df_master_details['Monthly_Allowance'].median(), inplace=True)
```

Task 3: Data Integration (Merging)

Finally, we want to map the students' academic grades to their detailed profiles.

- **Step 1:** Load the `student_grades.csv` file.
- **Step 2:** Perform an **Inner Join** between `df_master_details` and `df_grades` using `StudentID` as the primary key. Record the shape of this new dataframe. Observe that this dataset only contains students who have BOTH profile details and graded records.
- **Step 3:** Perform a **Left Join** using `df_master_details` as the main (left) table and `df_grades` as the right table. Record the shape. Notice how students who lack grade records will have NaNs in the `CGPA` and `Credits_Completed` columns.

```
# Example for Left Join
```

```
df_final = pd.merge(df_master_details, df_grades, on='StudentID', how='left')
```

Task 4: Final Export

Save your fully cleaned and left-joined master dataset to a new CSV file named

final_student_database.csv (remember to set index=False).